

LexGraph — Technical Report

Leakage, Discretisation and Authority Grounding in Indian Supreme Court Judgment Prediction

CSE 573: Semantic Web Mining — Arizona State University

Instructor: Prof. Hasan Davulcu TA: Anshul Trivedi

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Abstract

This is the single authoritative document for the project. It supersedes and replaces the specification (`PROJECT.md`), the chronological development log (`WORKLOG.md`) and the nine stage-acceptance reports (`reports/*.md`), all of which were removed; their content is folded in here and their history remains in git. A plain-language companion for readers without a background in law or machine learning, `LexGraph_Explained.tex`, sits in this same folder.

Two results are established. **First**, the input construction used by the published state of the art on this task beats a facts-only input by $+0.111$ to $+0.181$ AUROC, and essentially all of that gap is disposition language surviving the deletion of the operative order. The residue extends roughly **128 words** past the boundary — a figure found independently on four splits — and a word-count-matched window set back past it scores no better than the facts on any of them. Across 26,688 judgments the effect is significant in all eight decades from the 1950s to the 2020s. The court’s reasoning does carry real signal, but only about $+0.025$ AUROC: the apparent effect is four to seven times its true size.

Second, projecting extracted facts into a symbolic legal vocabulary costs 0.097 AUROC, which is most of what survives extraction. This is a property of discretisation rather than of any particular ontology: a corpus-induced vocabulary of equal granularity is statistically indistinguishable from the hand-authored one ($P = 0.846$), more labels do not help, and the cost is concentrated precisely where the vocabulary *succeeds* in naming a fact.

Neither critique is new to the literature, and §17 positions both against verified prior work. What is defensibly ours is methodological, and §17 says which parts. Three constructive results stand alongside: statute prediction at $1.9\times$ the strongest baseline, precedent retrieval at $7.4\times$ a popularity control, and a faithfulness-tested trace mechanism. Every number here is read from a file under `experiments/`.

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1 How this document is organised, and the specification section map

The project was built against a written specification whose section numbers (§5.2, §6.2, §10.1, ...) are cited in code comments across **50 source files**. That specification file has been removed in favour of this report, so those references are preserved here: each stage below carries its original spec number in the heading, and the table maps them.

Spec §	Stage	Here
3.1	Voyager/LLM environment	§2
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10.1	Outcome models and feature groups	§13
10.2	Ablations	§13
11	Traces and faithfulness	§14

Results that are not in the specification at all — the leakage measurement that became the project’s main finding — are in §15. §18 is the development record: the defects found and what each would have caused. §19 lists claims held and withdrawn. §20 is what is *not* established.

2 Environment spec §3.1

All model calls go to ASU’s hosted endpoint (Voyager, <https://openai.rc.asu.edu/v1>), through the single door in `src/llm/client.py`. No paid API is used anywhere.

- **No VPN required.** The endpoint resolved to private 10.139.64.x historically; as of 2026-09-18 it resolves to public Cloudflare addresses and answers from a residential connection.
- **Generation:** llama4-scout-17b by default. **Embeddings:** qwen3-embedding-8b (4096-dim). Only three of the roughly forty-nine advertised models serve /v1/embeddings; the rest 404 there, and the model list carries no capability metadata, so probing is the only way to tell an embedder from a generator.
- **Content-addressed cache**, keyed `sha256(model, messages, params)` in SQLite. Re-running an experiment is free — the LLM baseline ran at a 100% hit rate over 600 calls. The cache holds 162,230 entries.

- **ok=False means “never got an answer”**, which is deliberately distinct from a model that replied with nothing. Conflating the two turned a network outage into a finding twice before this was enforced (§18).
- **90-second timeout.** The client default of $600\text{ s} \times 3$ retries froze an early batch for over twenty minutes with no output.
- **Parallelism:** 96 workers for extraction, 24 for embeddings; throughput went from 10.3 to 38.6 documents per second.

Setup: `.venv/bin/pip install -r requirements.txt`, then `.env` at the project root with `VOYAGER_API_KEY=<key>` from `voyager.rc.asu.edu` → LLM Access → Create Key.

3 Corpus spec §4

26,688 Indian Supreme Court judgment PDFs, 1950–2025, 6.8 GB, held locally in `Data/raw_pdfs/<year>/`. A provided spreadsheet identifies **6,954** of them as land or property disputes; that subset is the working corpus for §4–§14, and §15.3 drops the restriction and uses all 26,688.

Extracting the archives. Two of the four source zips contain filenames with invalid UTF-8 bytes. APFS rejects those names, unzip misreports it as write error (disk full?), then blocks on a Continue? (y/n) prompt — and on non-interactive stdin takes EOF and **silently aborts the whole archive**, exiting 0. That is how a first attempt produced 12,422 of 26,688 files and reported success. Extraction must go through Python’s `zipfile` with each name sanitised (`encode("utf-8", "replace")`, NFC-normalise, strip control characters), and the count verified afterwards.

Joining spreadsheet rows to PDFs. `src/data/adapter.py` joins on the exact filename the spreadsheet supplies: **6,954 of 6,954 (100%)**, asserted at import time against `MIN_PDF_JOIN_RATE = 0.99` so a regression halts the pipeline instead of silently shrinking the dataset. An earlier normalised-title join achieved 0.6%, because judgment filenames carry a copy marker after the year (`..._on_23_May_1957_1`) and titles do not — so stripping “trailing digits” from both removes the *year* from every title and every pair mismatches by exactly that year.

Text. 6,954 of 6,954 extracted, zero failures, zero near-empty. Median 5,302 words.

What the dataset readme claims and the PDFs do not have. The readme describes structured sections. Sampled across six decades, FACTS appears as a header in **0** of 12 judgments, ISSUE in 2, ARGUMENT in 2, REASONING in 0. There is nothing to segment on, which is why §5 classifies units by cue rather than by heading.

Citation metadata. The provided “precedents” lists mix genuine case-to-case citations with statute and article references; only about **32.5%** are real case citations. The classifier output used downstream is `Data/processed/citations_classified.jsonl`. The metadata is Indian Kanoon’s link graph and is independently scraped, not ground truth — English and Privy Council authorities are systematically absent. Separately, only **11,991 of 46,904** case-citation edges (25.6%) point at a case that is in this corpus at all, because the corpus is Supreme Court only while judgments cite High Courts, the Privy Council and English decisions. Recall must therefore be reported against *reachable* authorities with the raw figure beside it.

4 Labels spec §5.1

Two independent labellers, made to agree.

Rules (`src/data/labels.py`) locate the operative order by regular expression and classify it. The hard part is that Indian orders dispose of several proceedings in one breath — “the appeal is allowed ... and the writ petition stands dismissed” — where the second clause is a *consequence* of the first and a naive matcher reads it as the outcome. Proceedings are therefore ranked into tiers: an appeal or special leave petition (`_PROC_T1`) governs, and a writ petition, suit, revision or application (`_PROC_T2`) is consulted only when tier 1 is silent. That change alone cut rules-versus-LLM conflicts from 196 to 107. A bounded gap (`_GAP`) forbids a proceeding noun from binding across another disposition verb, and a negator check rejects “we do not think it proper to again remand the matter”.

LLM sees the same order window and must quote the span it relied on.

Agreement: 87.0% on the 4,317 cases where both decided and the quote verified. Conflicts are held as UNRESOLVED and queued in `Data/interim/label_conflicts.jsonl` rather than resolved by trusting one labeller.

Label	Cases	Share
LOSE	2,964	42.6%
WIN	2,365	34.0%
PARTIAL	624	9.0%
UNRESOLVED	562	8.1%
UNKNOWN	289	4.2%
REMAND	116	1.7%
OTHER	34	0.5%
Total	6,954	

Table 1: The binary task uses WIN and LOSE only: 5,329 cases.

5 Masking spec §5.2

`src/data/mask.py` keeps only material that existed before the court decided. Units are classified by cue, not by heading, because there are no headings (§3).

Section	Kept	Why
Caption, parties, judges	no	Judge identity predicts but is not a legal signal; party names let a model memorise repeat litigants.
Facts	yes	The genuine input.
Procedural history	<i>isolated</i>	Legitimately known in advance but a shortcut, so it goes to its own channel prior_court_text and feature group Q, never mixed in silently.
Arguments of counsel	yes	Written by the parties, before the decision.
Court’s reasoning	no	Written after, knowing the answer.
Operative order	no	This is the label.
HEADNOTE (pre-2000)	no	An editorial summary that states outcome and authorities outright.
Page furniture	no	Indian Kanoon URLs, page numbers, running headers.

Within kept units there is a second, sentence-level scrub: sentences matching OUTCOME_CUE are dropped, and sentences matching PRIOR_COURT_DISPOSITION are diverted to the shortcut channel. Average retained length is about 2,500 words of the original 5,300.

Three variants are produced per case, which is what makes §15 possible: masked_text (facts and pleadings only), ildc_style_text (everything but the order) and ildc_style_no_headnote.

The sensitivity control, and why everything else depends on it. A claim to have removed leakage is untestable on its own. So a deliberately leaky arm is built — a model trained on *only* the deleted text — and must score near-perfectly. If it does not, the order detector is broken and the masked arm cannot be trusted either.

Input	Acc.	Macro-F1	AUROC
Majority class	0.528	0.345	0.500
Prior court’s disposition only	0.562	0.562	0.585
Masked text	0.629	0.626	0.674
Whole unmasked judgment	0.746	0.745	0.827
The removed order only	0.931	0.931	0.981

Table 2: Leakage probe, forum_heldout, $n_{\text{test}} = 902$. experiments/paper3/leakage_probe_forum_heldout.json. An LLM version agrees: 95.7% accuracy on the order alone, 62.0% on masked text ($n = 350$).

6 Splits spec §5.3

A random split fails here twice over. The same dispute generates several judgments (appeal, review, connected matter), so groups must not be split; and training on later cases to predict earlier ones is time travel.

- forum_heldout — three forums of origin held out entirely. Train 3,350 / dev 455 / test 902. Hash 6e346dd75b7e7d84.

- temporal_2004_2013 — train ≤ 2004 , dev 2004–2013, test > 2013 . Train 3,446 / dev 480 / test 768. Hash 43d9d645667ad3a5.

Every results file records the split hash, so a number traces to the exact split that produced it.

A conflict a unit test caught. Grouping by a case group’s *maximum* year put 1970 cases in “test > 2013 ”; by *minimum* year it put 2025 cases in “train ≤ 2004 ”. Both are time travel, and no assignment satisfies both constraints for a group straddling the boundary. The 13 straddling groups are dropped and the count recorded. tests/test_stage1.py enforces it.

7 Fact extraction spec §6.1

A model reads each segmented unit and proposes *facts*: atomic assertions, each carrying a verbatim quote, who asserted it, and whether it was disputed. src/grounding.py locates the quote in the source; a fact whose quote cannot be found is **discarded, not corrected**.

Grounding rate 0.92. Current scale: **96,623 facts across 4,702 cases**, median 17 per case, 15,377 claim records.

Segmentation. The first implementation split on blank lines and returned *one unit* for 57.7% of cases, because this PDF text frequently contains no blank lines. A sentence-window tier was added: median units per case 1 $\rightarrow$ 47, unusable cases 57.7% $\rightarrow$ 2.6%.

The verifier was the bug, not the extractor. An early version rejected 42% of quotes that were in fact correct: PDF extraction inserts hyphenation artifacts (plain- tiff’s), and a 20-character floor rejected true short quotes such as “Appeal dismissed.” (15 characters). After repairing hyphenation and lowering the floor, the verified rate on a hand-checked set went from 58.3% to 91.7%, with 15 of 16 confirmed present by eye.

Model selection. Six models on an identical 20-case set:

Model	Ground.	Leak	Attrib.	Facts/case	s/case
llama4-scout-17b	0.92	0.03	0.27	18.4	4.2
llama4-maverick-17b	0.93	0.01	0.41	23.5	28.7
qwen3-30b-a3b-instruct-2507	0.92	0.03	0.40	10.5	15.7
olmo3-32b-instruct	excluded — about 50 s per call				
glm-5-3-flash	excluded — hangs on long prompts				
glm-5-3	excluded — too slow at corpus scale				

Table 3: experiments/paper1/extractor_benchmark.json. maverick is better on every quality axis and 6.8 $\times$ slower; scout was chosen for corpus-scale runs and the trade-off is kept on the record.

Attribution, and an unexplained ceiling. Party attribution sits at **0.19–0.20**. The model records admittedness in disputed_status rather than asserted_by, and reading both raised coverage from 17% to 96% — but the *rate* at which a fact is attributed to a party has not moved. A prediction that recovering wrongly-deleted argument text would fix it was tested and refuted (§19). The cause is unknown.

8 Canonical vocabulary spec §6.2

Each fact is mapped from free text to a controlled label drawn from a 345-entry vocabulary generated from the ontology’s named fields. `ontology/ontology_v1.yaml` defines 15 claims, 66 elements, 6 defence families and 4 remedy tiers.

The appeal of a controlled vocabulary is comparability across cases: you can count element satisfaction, mine patterns, and reason symbolically. The cost is that two distinct facts mapping to one label become indistinguishable — and §16 shows that cost is severe.

Three failures worth not repeating. (i) The canonicaliser’s self-reported “NEW” rate of 0.7% was untrustworthy: 16.7% of facts were out of vocabulary and unflagged. Out-of-vocabulary is now decided by checking the answer, not the flag; at scale the model flags only 75 of 3,155 (2.4%). (ii) Batching with a shared union menu wrecked label fidelity, and out-of-vocabulary rose with menu size (51%/75%/76% at batch 4/8/16); per-fact menus restored parity (32.3% vs 34.4% unbatched). (iii) Repointing `PROMPT_ID` left `canonicalize_one` validating a batched reply against a single-object schema, so it returned `None` for everything — and it is the *fallback* path. Split into `PROMPT_ID_ONE` and `PROMPT_ID`.

9 Claim families spec §7

Clustering 7,441 claims (HDBSCAN and agglomerative, `src/cluster/`) produced six clusters with a **silhouette of 0.0245** and 71.5% of points classed as outliers; NMI 0.237, ARI 0.083. A silhouette near zero means the clusters are not separated. Merging with a minimum family size let one family (`PrivateTitlePossession`) absorb 2,365 cases and nearly every other label.

The taxonomy is not supported by the data, and is reported as such rather than presented as a finding.

10 Fact patterns spec §8

FP-Growth over closed itemsets within each family, chi-square tested, Benjamini–Hochberg corrected, with bootstrap stability over 20 resamples.

Closed itemsets mined	498
Stable across $\geq 80\%$ of 20 bootstraps	311
Patterns statistically tested	351
Pass uncorrected $p < 0.05$	10
<i>Expected by chance alone</i>	17.6
Survive Benjamini–Hochberg	0

Ten patterns passed an uncorrected test — *fewer* than the 17.6 chance predicts. Without the correction those ten would have shipped as a pattern library. This is the clearest case in the project of a control changing the conclusion, and it follows directly from §16: pattern mining over symbols that carry no signal cannot find anything. The specification’s §8 deliverable is therefore **vacuous rather than met**.

11 Statutes spec §9.1, 9.2

src/authorities/normalize.py reduces citations to canonical CODE:provision form over a registry of 42 Acts. Three properties of real Indian citations forced the design:

- **Act names arrive truncated.** The most frequent “Act” in 1,200 sampled judgments is *Property Act* (644 mentions) — the *Transfer of Property Act* with its head cut off by a line break. *India Act* (516) is the *Government of India Act*. Truncations resolve through a path tagged `resolved_by="truncated"`, so a guess is never reported as a full-name match.
- **Generic back-references dominate:** *the Act, the said Act* (311+), which resolve only by looking back to the last Act actually named, within a bounded window.
- **Implicit Acts:** a bare *Article* means the Constitution; an *Order* or *Rule* means the Civil Procedure Code.

Coverage over 1,500 judgments: 58,295 citations, **61.4% resolved**, median 6 distinct provisions per case, 149 cases with none.

A copy control killed the first result. Predicting the provisions that apply reached micro-F1 0.447 and looked strong. A system that does nothing but copy provisions already named verbatim in the input scores **0.713** — far higher. The “prediction” was largely string matching. The honest version removes every provision the input mentions, leaving only those that must be inferred:

System	Micro-F1	P@1	R@10
Global top-5 prior	0.069	0.134	0.418
Claim-family prior (<i>strongest baseline</i>)	0.093	0.173	0.503
Copy from text (<i>control, now at floor</i>)	0.000	0.081	0.385
Masked text	0.173	0.236	0.660
Extracted fact text	0.159	0.206	0.581
Canonical atoms	0.109	0.161	0.464

Table 4: Novel provisions only. 46 labels, $n_{\text{test}} = 702$. experiments/paper2/statute_prediction_forum_heldout_labels_novel.json.

Micro-F1 0.173 against 0.093 — $1.9\times$ **the strongest baseline**, with the copy control at zero by construction. A real positive result, and note the ordering recurs: raw text > extracted facts > canonical atoms.

12 Precedents spec §9.3

150 queries. Recall is against the **1,035 of 2,113** cited authorities (49.0%) that are themselves in this corpus (§3); the raw figure is stated beside it. Every retrieved authority is asserted to pre-date the query judgment — checked per result, not trusted to a filter.

System	R@10	R@50	MRR	nDCG@10
Mention baseline	0.022	0.112	0.037	0.014
Popularity (<i>control</i>)	0.032	0.103	0.072	0.032
Canonical atom overlap	0.021	0.058	0.046	0.021
Dense, mean-pooled	0.053	0.130	0.114	0.054
Dense, max-pooled	0.039	0.104	0.097	0.044
BM25	0.207	0.350	0.431	0.225
Dense over masked text	0.229	0.390	0.380	0.221
RRF(BM25 + dense masked)	0.238	0.453	0.428	0.243

Table 5: experiments/paper2/precedent_retrieval_forum_heldout.json.

R@10 0.238 and MRR 0.431 against a popularity control of 0.032 and 0.072 — about $7.4\times$ on R@10. Because the control is popularity, beating it by that margin establishes the system responds to the facts rather than exploiting citation skew.

Two sub-results: canonical atom overlap (0.021) falls *below* the popularity control, so the symbolic representation fails here too; and max-pooling made the dense arm *worse* (0.039 vs 0.053), a prediction that had to be withdrawn.

13 Outcome models spec §10.1, 10.2

Six explicitly separated feature groups, so any claim about what drives a prediction is testable by ablation. 1,415 features: F canonical fact atoms 1,313; P mined patterns 3; E element satisfaction **0 (empty, see §20)**; S predicted statutes 49; R retrieved precedents 6; C case metadata 38; Q prior court’s disposition 6.

Model	Acc.	Macro-F1	95% CI	ECE
Majority (global)	0.530	0.346	[0.332, 0.360]	—
Majority (per decade)	0.571	0.557	[0.527, 0.589]	—
1r F (facts only)	0.501	0.499	[0.464, 0.530]	0.270
1r F+P+E+S+R+C	0.513	0.513	[0.479, 0.546]	0.285
1r S only	0.599	0.599	[0.567, 0.629]	0.041
1r Q only	0.525	0.509	[0.474, 0.540]	0.034
gbm C only (metadata)	0.583	0.582	[0.550, 0.613]	0.091
gbm F+P+E+S+R+C	0.598	0.597	[0.566, 0.630]	0.107

Table 6: $n_{\text{test}} = 881$, 24 models run. experiments/paper3/outcome_forum_heldout.json.

Three uncomfortable observations. **Every confidence interval spans about 0.11 and they all overlap** — the best structured model (0.597) and a per-decade majority baseline (0.557) are not distinguishable at this sample size. **Adding facts to metadata does not help:** gbm C only reaches 0.582 on year and forum alone, while facts alone reach 0.517, and McNemar confirms the metadata-and-authority groups genuinely beat the fact group ($P = 0.0008$ for LR, $P = 0.004$ for GBM) — a real difference in the wrong direction for the project’s original thesis. And **calibration and accuracy come apart:** 1r S only is the best-calibrated arm at competitive accuracy.

LLMs do no better. Five configurations on the same 300 cases: LLM-0 0.491 macro-F1, LLM-FS 0.522, LLM-CoT 0.582, LLM-CF 0.469, LLM-CF+A 0.503, against structured GBM 0.599 and TF-IDF on masked text 0.593. Chain-of-thought’s edge over zero-shot is not significant (McNemar $P = 0.926$). Nothing clears a per-decade majority baseline by a resolvable margin, so the ceiling is not an artifact of a weak model family.

The ceiling looks intrinsic. 354 errors on 881 cases were categorised with a **base-rate control**, and it is decisive. Against an overall error rate of 0.402 the lifts are `family_unassigned` 1.12, `statute_mismatch` 0.99, `discretionary` 0.93, `extraction_thin` 0.88, `atoms_only_generic` 0.62. **Every lift is approximately 1.0** — not one of the pipeline’s own quality signals predicts where it will fail. Reported as raw counts (“175 errors involve a statute mismatch”) they would have read as a diagnosis and a work plan.

The stronger test: on 300 shared cases the structured model made 119 errors and LLM-CoT 125; under independence one expects $119 \times 125 / 300 = 49.6$ wrong for both. Observed: **46**. The two systems’ errors are statistically independent, so they are not failing for a shared, fixable reason.

14 Traces spec §11

A deletion test. Remove the facts the trace cites, recompute features *from scratch*, and measure how far the predicted probability moves; compare against deleting the same number of random facts, and the facts the trace ranked least important. Deletion is applied to the **facts**, not the feature vector — deleting a feature the model weights heavily would be circular.

Facts deleted	Mean Δp	vs cited	95% CI
Cited by the trace	0.0650	—	—
Random	0.0209	+0.0441	[+0.0245, +0.0659]
Inverse-ranked	0.0180	+0.0470	[+0.0291, +0.0666]

Table 7: 45 cases. Cited facts move the prediction $3.1 \times$ as much as random, winning on 77.8% of cases. `experiments/paper3/trace_faithfulness_forum_heldout_gbm.json`.

Verdict: faithful. Faithful is not correct — the test shows the explanation describes the model’s actual computation, not that the computation is legally sound. Given that the same model predicts near chance, a faithful explanation of a near-chance prediction is an honest account of a weak inference.

15 The leakage result

This is not in the specification. It became the project’s main finding.

15.1 The construction

Six input constructions on **identical cases, split, labels and classifier** (TF-IDF into logistic regression). Only the text the model may read varies.

Input	Words	Acc.	Macro-F1	AUROC
order_only (<i>control</i>)	427	0.921	0.920	0.976
full (<i>ceiling</i>)	5,302	0.720	0.718	0.788
ildd_style (<i>all but the order</i>)	5,051	0.680	0.678	0.734
ildd_style_no_headnote	4,481	0.678	0.676	0.735
last_512_tokens	511	0.776	0.775	0.845
masked (<i>facts + arguments</i>)	2,501	0.621	0.619	0.668

Table 8: 895 test cases, 3,324 training. experiments/paper1/reasoning_ablation_forum_heldout.json. Paired bootstrap: last_512-masked = +0.1764 [+0.1480, +0.2064]; ildd_style-masked = +0.0660 [+0.0539, +0.0789].

last_512_tokens reaches **accuracy 0.776**; the published best on this task is **78%**. A plain TF-IDF model lands within half a point of a transformer result once it reads the same window. Note also that the winning arm reads **511** words against masked’s **2,501** — so “less text” cannot explain the gap; it runs hard the other way.

15.2 Where the advantage actually comes from

The final 512 tokens sit exactly where the operative order used to be. If the boundary is even slightly imperfect, that arm — and no other — is contaminated. Three measurements, every arm exactly 512 words so volume is held constant.

1. The setback curve. Four non-overlapping windows stepping back from the end. Both predicted shapes were written into the module before it ran: *residual leakage* predicts a cliff after the first window, because contamination cannot reach back 512 words; *reasoning* predicts gradual decay with every window above masked.

It is unambiguously the cliff (table in §15.3, all four splits). **512 words of the court’s reasoning taken from anywhere but the very end carry no more outcome signal than the facts do**, and on the corpus-wide runs slightly less.

2. The mechanism, visible in the text. Indian judgments announce the conclusion one sentence *before* the operative order:

“For the foregoing reasons we find no merit in the appeal. *The appeal is dismissed with costs.*”

The detector marks the boundary at the second sentence, which is **correct** — that is the operative order. The first sentence stays, and states the outcome just as plainly.

3. Depth. Sweeping the window back in fine steps locates the boundary, and AUROC falls together with the disposition-cue rate — what a single shared cause looks like. On forum_heldout: offset 0 gives AUROC 0.838 at cue rate 0.419; 32 words 0.780/0.296; 64 words 0.719/0.208; **128 words 0.684/0.131, no longer significant**; 256 words 0.676/0.099; 512 words 0.669/0.089.

What cue-scrubbing cannot do. Deleting every *cue-bearing sentence* from the final window drops it $0.838 \rightarrow 0.775$; evaluating only on the 489 cases with no detected cue gives 0.742 against masked’s 0.646 on those same cases. Both remain above masked, so sentence-level scrubbing does **not** fully clean the window — there is telegraphing language the patterns miss (“we find no merit in the submission”, “the conclusion is inescapable”). This is exactly why the setback curve is the primary evidence: it never relies on detecting a cue.

15.3 Generalisation: 26,688 judgments, every case type

Two objections survive the above — it might be a land-dispute artifact, and 842 test cases is not many. The intended answer was a replication on ILDC (34,816 cases), whose release is gated behind a review queue that may never be processed.

It proved unnecessary, because **the restriction to land disputes was ours, not the data’s**. All 26,688 judgments were already on disk and ILDC is drawn from the same population (Indian Kanoon’s Supreme Court judgments). `src/data/corpus_wide.py` extracts, rules-labels and masks every one in parallel, API-free: **2.8 minutes**, yielding **14,168 usable WIN/LOSE cases** (12,067 usable in every arm) across criminal, tax, constitutional, service, company and land matters.

This is a *generalisation test, not a replication*: it reproduces neither ILDC’s labels nor ILDC’s deletion boundary.

	land only		all case types	
	forum (842)	temporal (722)	temporal (2,288)	random (2,414)
masked	0.656	0.577	0.628	0.723
w0 (last 512)	0.838	0.716	0.739	0.862
w1	0.669	0.589	0.606	0.717
w2	0.647	0.577	0.616	0.715
w3	0.638	0.578	0.603	0.701
w0–masked	+0.181	+0.139	+0.111	+0.139
w1–masked	+0.013	+0.012	−0.022	−0.006
<i>significant?</i>	<i>no</i>	<i>no</i>	<i>no</i>	<i>no</i>
Advantage gone at	128 w	128 w	128 w	128 w

Table 9: Four independent splits. `leakage_setback_*.json`, `corpus_scale_setback_*.json`.

Not a pre-2000 reporting artifact. The pre-2000 HEADNOTE convention is a plausible alternative cause. It is not the cause — w0–masked is significant in **every decade**:

decade	1950s	1960s	1970s	1980s	1990s	2000s	2010s	2020s
<i>n</i>	112	438	377	292	278	338	349	230
Δ	+0.179	+0.325	+0.186	+0.180	+0.171	+0.163	+0.096	+0.115
sig.	yes	yes	yes	yes	yes	yes	yes	yes

The 1960s peak and mild post-2000 decline are consistent with the HEADNOTE convention *adding* to the effect, but the effect does not depend on it.

The reasoning’s real contribution. Cutting the contaminated tail off the whole-document arm isolates it:

Run	ilddc_minus128-masked	Sig.
land forum (842)	+0.027 [+0.014, +0.039]	yes
land temporal (722)	+0.022 [−0.0002, +0.043]	no
corpus temporal (2,288)	+0.023 [+0.010, +0.035]	yes
corpus random (2,414)	+0.029 [+0.022, +0.036]	yes

Four point estimates between +0.022 and +0.029, significant in three; the one failure is the smallest test set, so the earlier non-replication was a power problem, not a contradiction.

15.4 Is the baseline itself clean?

The correction above came from assuming an arm was clean without testing it, so the same question must be put to masked — the baseline every comparison runs against and the text extraction reads. There was a concrete reason for concern: this repository contains *two* disposition detectors that disagree by construction. `mask.py`’s `OUTCOME_CUE` requires the proceeding noun *adjacent* to the verb; `labels.py`’s patterns allow a 90-character gap and also cover “the appeal fails” and “we hereby dismiss the appeal”.

Arm	AUROC	Δ	Sig.
masked (2,647 words)	0.662	—	—
masked_strict (<i>escaped sentences removed</i>)	0.662 −0.0001 [−0.0007, +0.0005]		no
masked_w0 (<i>its last 512 words</i>)	0.671	—	—
masked_w1 (<i>set back, volume-matched</i>)	0.644 +0.027 [−0.002, +0.056]		no

Table 10: experiments/paper1/masked_integrity_forum_heldout.json.

Sound. Only 109 sentences across 839 cases escape — **one word per case** out of 2,647 — and removing all of them changes AUROC by −0.0001. Reading the list explains why they are harmless: statutory quotations (“that a suit must fail by reason of some formal defect”, Order 23 CPC), prior-court dispositions whose forum the narrow regex did not name (“A single judge partly allowed the appeal”), legal propositions and conditionals. The same setback design applied to masked gives +0.027 with an interval spanning zero at cue rates 0.054/0.031, against +0.169 [+0.136, +0.202] at 0.419/0.089 for the contaminated window.

A caveat the test supplied about itself. The naive check — simply deleting masked’s last 128 words — *was* significant (+0.0058 [+0.0023, +0.0092]). That is confounded: deleting 128 words of anything loses information. Only the volume-matched arms distinguish contamination from ordinary loss, and without them this section would have reported a leak that is not there.

The headline

The input construction used by the published state of the art beats a facts-only input by +0.111 to +0.181 AUROC across four splits. **Essentially all of that is an artifact of imperfect order removal.** Deleting the operative order leaves roughly **128 words** of outcome-telegraphing text in place — a boundary found independently four times — and a word-count-matched window set back past it scores no better than the facts on any split.

With the tail removed, the court’s *entire* reasoning adds about +0.025 AUROC. The apparent effect is **four to seven times** its real size. Measured on **26,688** Supreme Court judgments, 1950–2025, every case type, significant in **all eight decades**.

16 The discretisation result

A **representation ladder**: identical classifier, split, labels and 881 test cases at every rung, with only the representation changing.

Rung	Representation	Features	Macro-F1	AUROC
0	majority class	—	0.346	—
1	masked judgment text	50,000	0.608	0.657
2	extracted fact text	41,142	0.586	0.614
3	canonical fact atoms	1,313	0.505	0.517

Table 11: experiments/paper1/representation_ladder_forum_heldout.json.

Extraction costs **0.043**. Canonicalisation costs a further **0.097**, landing at 0.517 — essentially chance. Five independent tests agree:

1. **The ladder:** 0.657 → 0.614 → 0.517.
2. **Is it *this* ontology?** No. Raw text beats both an authored and a corpus-induced vocabulary decisively, and the two vocabularies are indistinguishable from each other:

Comparison	ΔAUROC	95% CI	<i>P</i>
text – induced	+0.1138	[+0.0732, +0.1545]	0.000
text – ontology	+0.0887	[+0.0430, +0.1352]	0.000
induced – ontology	−0.0251	[−0.0711, +0.0200]	0.846

3. **Would more labels help?** No. A sweep from 345 to 1,000 entries does not recover the gap, and a frequency-threshold sweep sits at chance throughout.
4. **Where does the loss occur?** The sharpest test. Facts were split into those the vocabulary *can* name and those it cannot: the cost of naming is **0.087** in group A (text 0.544 vs atoms 0.457) and **0.002** in group B (0.526 vs 0.524). The cost is concentrated *where the vocabulary succeeds*, which rules out “coverage is incomplete, fix the vocabulary”. Small samples ($n = 143$ and 127); group A’s interval spans zero at $P = 0.083$, so this line is suggestive on its own.
5. **A pre-registered prediction:** an LLM given canonical facts would do worse than the same LLM given raw text. It did: macro-F1 0.469 vs 0.491.

The downstream consequence is §10: 0 of 498 patterns survive correction.

17 Prior art, and what is actually ours

Both headline findings independently rediscover active, recently-published work. Three papers were located and **verified by fetching the abstract**, not taken from a search summary.

Our finding	Prior art
Facts-only $\ll$ last-512 on Indian SC judgments	Nigam & Deroy , <i>Fact-based Court Judgment Prediction</i> , arXiv:2311.13350 (Nov 2023). Runs facts-only on <i>ILDC itself</i> and reports it worse than the full-document setup. Our direction, on the actual benchmark, three years earlier.
The gap is leakage / short-cut learning	Watson et al. , arXiv:2607.04261 (Jul 2026). 33,158 UK Employment Tribunal claims, nine authors. Identifies leakage by <i>human judgment stratification</i> and lexical feature flagging; trains a leakage-only model on 200 trigrams.
Ontologising legal text destroys downstream signal	Sadowski & Chudziak , arXiv:2608.12326 (May 2026). Six LLMs, three ontology-learning methods.

Unverified leads, to check before citing: ECtHR work reporting that masking verdict words was “not completely successful”; *Multi-Legal-Bench* on label leakage (arXiv:2605.29738); Medvedeva et al., *Rethinking the field of automatic prediction of court decisions* (Springer AI & Law 2021).

17.1 What survives as ours

1. **The volume-controlled positional setback curve.** Every prior method located identifies leakage *lexically* or by *human annotation*. Ours needs no cue detector, no annotator and no feature flagging — only word-count-matched windows at varying distance.
2. **Evidence that this matters, not merely differs.** Our own cue audit **failed to replicate in three of four runs** (+0.150, −0.058, +0.018, ≈ 0) while the positional curve held in all four with the same boundary. So the claim is not “we have a different method” but *the method the field uses to localise leakage is unstable across case mixes, and here is a cue-free one that is not* — shown on 26,688 judgments with a failed lexical arm inside every run.
3. **The ~ 128 -word margin**, found four times. No located work states a distance. This is the actionable output: a benchmark built this way needs that margin, not zero. It is a lower bound under one baseline, not a constant — against `ildc_facts_only` rather than masked the substitute run put it at 512 with 128 and 256 marginal, so quote the sweep table and name the baseline.
4. **The induced-versus-authored control.** Sadowski & Chudziak explicitly conclude their loss is “*not simply from vocabulary discretization alone*” and do not compare an induced vocabulary against a hand-authored one. We do, at $P = 0.846$. **This is the gap in our favour and the less crowded area**; it should lead the paper.
5. **A cleared baseline** (§15.4). Prior work does not verify its own clean arm with the same instrument.

17.2 Are we “breaking the industry”? No

- **The subfield already knows.** Three verified papers across 2023–2026 cover both directions; one is three months old.
- **Deployed outcome prediction cannot have this bug.** Litigation analytics predicts from judge, court, party and case-type features — it *cannot* read the court’s reasoning, because at prediction time nobody has written it. The leakage is purely an artifact of retrospectively feeding decided judgments to a model: a **benchmark-construction finding, not an industry one**.
- **Scale limits the claims.** Every outcome-model CI spans about 0.11.

What this is instead: a well-controlled workshop paper (NLLP at EMNLP, or JURIX) with two real methodological contributions and a constructive positive result. A good outcome for a class project; not a disruption.

18 Development record: the defects that mattered

Most of the effort went not into building the pipeline but into establishing that its numbers meant what they appeared to mean. Each defect below would have produced a confident, publishable, wrong answer.

Defect	What it would have caused	How it was caught
unzip silently aborting two archives on invalid UTF-8 filenames	12,422 of 26,688 judgments, exit code 0	Counting the result
Normalised-title PDF join stripping the year from every title	A 0.6% join rate presented as a corpus	An asserted join-rate floor
Segmentation returning one unit for 57.7% of cases	Extraction silently operating on 2.6% of its input	Median-units sanity check
Quote verifier rejecting 42% of <i>correct</i> quotes	Grounding rate understated; good extractions discarded	Hand audit of 16 quotes
Consequential dismissals read as the disposition	196 cases labelled with the wrong outcome	Auditing rules-vs-LLM conflicts
Group-whole and time-respect split constraints in conflict	Training on the future to predict the past	A unit test
Attribution reading only asserted_by (84% court_narrative)	Coverage reported as 17% instead of 96%	Inspecting the field distribution
Canonicaliser’s self-reported “NEW” rate of 0.7%	A vocabulary declared complete while 16.7% was unnameable	Checking answers, not the flag
Batching with a shared label menu	Label fidelity silently halved	Batch-size sweep (4/8/16)

Defect	What it would have caused	How it was caught
canonicalize_one dead after a prompt-ID change	The <i>fallback</i> path returning None for everything	Reading the validation path
Driver running two stages at once, only the last exit code checked	A failed stage advancing the chain as success	A 26-minute orphaned process
macOS idle sleep mistaken for a hang (78, 194, 232 min)	Three runs killed for no reason	pmset -g log
Burden annotation 92% one class, one value never used	Legal metadata that was a constant, read downstream as fact	A degeneracy gate that <i>refuses</i>
Order-window detector swallowing whole short documents (59 cases)	A third of all unusable cases, cause unknown	Unusable-case audit
Caption detector eating argument text	545 of 5,255 units — about 1.2M words corpus-wide — deleted as “caption”	Reading what was dropped
Loop-variable shadowing in the metrics dump	Metrics overwritten by a loop counter	Implausible output value
Per-item LLM loop with no retries (twice)	An outage reported as “the corpus does not state these elements”	ok=False ≠ empty answer
A closing summary hard-coding “On ILDC’s own data”	A substitute result labelled as a real one	The smoke test

18.1 Five transferable lessons

1. **A control is worth more than a method.** The copy control killed a statute result (0.447 → an honest 0.173); the popularity control validated a precedent result; the base-rate control dissolved an entire error taxonomy; Benjamini–Hochberg stopped a pattern library shipping; volume-matching stopped a non-existent leak being reported. In every case the control changed the conclusion.
2. **An indirect argument that an objection is unlikely is not a measurement.** The last-512 contamination was dismissed on two true-but-unresponsive grounds before being tested directly. A detector can mark a boundary *correctly* and still leave the preceding sentence.
3. **Distinguish “no answer” from “the answer is nothing”.** This single confusion produced two separate false findings.
4. **Assert your joins.** A pipeline that silently processes 12% of its input is worse than one that crashes.
5. **Elapsed time is not progress, and 0% CPU is not a hang.**

19 Claims held and withdrawn

Belief	What the measurement said
“The last-512 window is clean, so its advantage shows the court’s <i>reasoning</i> carries the signal.”	Wrong, and this project asserted it before testing it. The advantage vanishes once the window is set back 128 words, on all four splits. Essentially all of it is residual disposition language; the reasoning’s own contribution is about +0.025.
“Extraction preserves the signal in the text.”	Loses 0.043 AUROC; canonicalisation a further 0.097.
“The ontology’s vocabulary is the problem; an induced one will do better.”	Indistinguishable: -0.025 , $P = 0.846$. The loss is discretisation itself.
“The procedural shortcut dominates everything.”	Weak. Q only reaches AUROC 0.498 — chance.
“Max-pooling will fix the dense retrieval arm.”	Worse: R@10 0.039 vs 0.053. The deficit is in the facts.
“ILDC leaks the operative order.”	It does not. The authors explicitly delete it, and we verified the claim. What it retains is the reasoning — a different and defensible choice.
“Recovering the 1.2M words of wrongly-deleted argument text will improve attribution.”	Flat: $0.20 \rightarrow 0.19$. Absolute attributed facts rose about 15%; the rate did not move.
“A better grounding model will recover the missing Fact and Issue concepts.”	Tested across five models. Rules learned under one fire on nothing under another, and the recovered rules were single-example artifacts generalising to zero real sentences.
“A bigger model will produce usable element-burden metadata.”	Two models failed the same degeneracy gate; the larger, more instruction-disciplined one also failed.

The last two illustrate a pattern worth naming: on this project **model capability was almost never the binding constraint**. Instruction discipline, evaluation design and data coverage were.

20 What is not established, and open blockers

B2 — element burden metadata does not exist spec §8.3. The ontology requires every element to record who bears the burden of proof, to what standard, and when it shifts, and supplies this for **0 of 66**. Two models produced degenerate output (92% and 87% assigned to one class; one standard never used at all, including where it unambiguously applies). A gate refuses to save such annotations, so **feature group E is empty** and every result labelled F+P+E+... contains no E. The fix is a few hours of expert hand annotation (`Data/gold/element_burden_sheet.csv`), not a larger model.

B3 — the vocabulary cannot be frozen spec §6.2. 20.3% of 79,496 canonical facts are out of vocabulary against a 5% freeze threshold. Note §16 shows improving coverage would not rescue the predictive result.

B4 — no gold annotation set spec §6.4. The specification asks for ≥ 150 cases annotated by a legal reviewer with agreement on a 30-case overlap. The **150 cases are prepared** (`Data/gold/gold_tasks.json`) and an annotation tool exists (`src/annotate/app.py`), but no reviewer has annotated them, so **extraction precision and recall against gold cannot be computed**. The four proxies in use and what each does *not* establish: grounding rate 0.92 bounds fabrication but says nothing about recall; the legal-statement filter bounds one precision failure mode; rules-vs-LLM agreement 87.0% estimates outcome-label noise, not fact quality; the representation ladder measures downstream utility but cannot attribute a loss to a specific extraction error. **This is the single biggest gap for a reviewer, and it is blocked on human time, not compute.**

B5 — the ILDC replication (optional). The licence request sits in a review queue expected not to be processed. `src/eval/ildc_ablation.py` is written, setback-armed and verified end to end via `-smoke-test`, and runs the moment `Data/ildc/` is populated. It is **no longer load-bearing**: §15.3 covers the same population at comparable scale. What ILDC would still add is gold labels and *its own* deletion boundary, so it could show that *their* specific cut leaves residue rather than that a cut *like* theirs does. The module does not route around the gate. Note the authors are **not** inactive — Nigam published through 2026 and Modi is organising the JUST NLP 2025 workshop — so a direct email beats the queue.

Other open items.

- **What the cue scrub cannot separate.** `w0_scrubbed` (0.775) and `w0_cue_free` (0.742) both stay above masked, so some telegraphing language escapes the patterns. Whether that remainder is undetected leakage or genuine summative reasoning is unresolved; the setback curve sidesteps the question rather than answering it.
- **Human verification of outcome labels spec §5.1.** The check of 200 labels is outstanding; the 562 UNRESOLVED conflicts are the highest-information place to start.
- **The attribution ceiling** (§7). Cause unknown.
- **Statistical power.** Every outcome-model CI spans about 0.11; the structured model and a per-decade majority baseline cannot be separated at $n = 881$.
- **§8 is mis-specified for this corpus**, not merely unmet.
- **Not started:** the demo dashboard; the team roster and the official proposal template.

21 Reproduction

1. `.venv/bin/pip install -r requirements.txt`; create `.env` with `VOYAGER_API_KEY=<key>`.
2. `./run_downstream.sh` — re-executes itself under `caffeinate -dimsu` so idle sleep cannot stall it, aborts on any stage's non-zero exit, and runs a QA gate after extraction. Every stage is resumable. `make help` lists individual targets.
3. Headline experiments:

```
python -m src.eval.leakage_probe -split forum_heldout
python -m src.eval.reasoning_ablation -split forum_heldout
python -m src.eval.leakage_setback -split forum_heldout
```

```
python -m src.eval.masked_integrity -split forum_heldout
python -m src.data.corpus_wide
python -m src.eval.corpus_scale_setback -split temporal
python -m src.eval.representation_ladder -split forum_heldout
python -m src.eval.vocab_ab -split forum_heldout
python -m src.authorities.statute_predict -split forum_heldout
-rebuild-targets
python -m src.authorities.precedent_retrieve -split forum_heldout
python -m src.predict.models -split forum_heldout
python -m src.trace.evaluate -split forum_heldout -model gbm
```

4. Every number in this report is read from a file under `experiments/.src/report/figures.py` reads the same files and *raises* rather than substituting a default if one is missing, so a stale figure cannot be produced silently.

22 Repository map

Path	Contents
<code>writeup/</code>	This report and <code>LexGraph_Explained.tex</code> , the plain-language companion. The only documentation in the repository.
<code>src/data/</code>	Adapter, labels, masking, segmentation, splits, ontology conversion, <code>corpus_wide.py</code> .
<code>src/extract/</code>	Fact and claim extraction, canonicalisation, vocabulary.
<code>src/eval/</code>	The experiments: leakage probe, reasoning ablation, setback curve, masked integrity, corpus-scale setback, representation ladder, vocabulary A/B and diagnosis, error analysis, ILDC replication.
<code>src/authorities/</code>	Statute normalisation and prediction, precedent retrieval.
<code>src/predict/</code>	Features, outcome models, LLM baselines.
<code>src/trace/</code>	Trace construction and the faithfulness deletion test.
<code>src/llm/client.py</code>	The single door to the model endpoint: cache, timeout, retries.
<code>src/grounding.py</code>	Verbatim quote location with hyphenation repair.
<code>prompts/</code>	Versioned prompt templates, keyed by the IDs the cache records.
<code>ontology/</code>	<code>ontology_v1.yaml</code> and the vocabulary proposals.
<code>experiments/</code>	Every result file. Nothing in this report is sourced elsewhere.
<code>tests/</code>	Unit tests, including the split-constraint test that caught a time-travel bug.
<code>Data/interim/</code>	Derived artifacts; large ones are gitignored and rebuild deterministically.
<code>Data/gold/</code>	The 150 prepared gold cases and the 66-element burden sheet (B2, B4).
<code>Data/raw_pdfs/</code>	All 26,688 judgment PDFs (gitignored, 6.8 GB).
<code>run_downstream.sh, Makefile</code>	The driver and individual targets.

This report replaces `PROJECT.md`, `WORKLOG.md` and `reports/*.md`, whose history remains in git. Every figure was read from a results file in `experiments/`.